# Learning notes: Gaussian measures and the equivalence of SPDE priors

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These notes go slowly through the mathematics of equivalent and singular Gaussian measures, with SPDE priors in mind. Every figure is produced by a Jupyter notebook in this folder, so each statement can be checked by running code. Chapter 1 pairs with `01_feldman_hajek.ipynb`.

## Contents

**1 Feldman–Hájek: when are two Gaussian priors "the same kind of thing"? 1**
1.1 Equivalent and singular measures . . . 1
1.2 Infinitely many independent coordinates . . . 2
1.3 Mean shifts and the Cameron–Martin space . . . 3
1.4 The Feldman–Hájek theorem . . . 5
1.5 Random functions: range versus amplitude . . . 6
1.6 Take-aways . . . 7

# 1 Feldman–Hájek: when are two Gaussian priors "the same kind of thing"?

**Why we care.** Take a reference prior $P$, the stationary SPDE $(\kappa_0^2 - \Delta)u = \xi$, and an alternative $Q$ that differs from it in some way: a different parameter, an added source, a changed coefficient. Whenever we compare, learn, or sample, we implicitly use the density ratio $dQ/dP$: likelihood ratios, KL divergences, MCMC acceptance ratios, "correct the reference by a few directions". On a fixed mesh this ratio always exists. The question is whether it *survives mesh refinement*. If it does not, every one of those quantities blows up as the mesh gets finer, and no fixed-size correction can repair the reference. The Feldman–Hájek theorem tells us exactly when this happens.

*Intuition.* Two Gaussian measures are equivalent if and only if a single, infinitely resolved sample cannot tell you which of the two produced it. If one perfect sample *can* tell them apart, they are mutually singular.

Everything below builds toward this sentence and then checks it with numbers.

## 1.1 Equivalent and singular measures

**Definition 1.1.** Let $P, Q$ be probability measures on the same space.

- $Q$ is *absolutely continuous* w.r.t. $P$ ($Q \ll P$) if $P(E) = 0 \Rightarrow Q(E) = 0$ for every event $E$.

- $P$ and $Q$ are *equivalent* ($P \sim Q$) if $Q \ll P$ and $P \ll Q$: they have the same impossible events.
- $P$ and $Q$ are *mutually singular* ($P \perp Q$) if there is an event $E$ with $P(E) = 1$ and $Q(E) = 0$: they live on disjoint sets.

By the Radon–Nikodym theorem, $Q \ll P$ exactly when there is a density $\frac{dQ}{dP}$ with $Q(E) = \int_E \frac{dQ}{dP}\, dP$. For equivalent measures the density is finite and positive ($P$-almost surely).

**Finite dimensions: equivalence is the default.** $P = \mathcal{N}(0, 1)$ and $Q = \mathcal{N}(0, 1.5^2)$ have

$$\frac{dQ}{dP}(x) = \frac{1}{1.5} \exp\Big(\frac{x^2}{2}\big(1 - 1.5^{-2}\big)\Big),$$

which is finite and positive for every $x$ (Fig. 1b). The only way two Gaussians on $\mathbb{R}^n$ can be singular is if one is degenerate and lives on a lower-dimensional subspace (Fig. 1c). In general, two *non-degenerate* Gaussians on $\mathbb{R}^n$ are always equivalent, however different their means and covariances are.

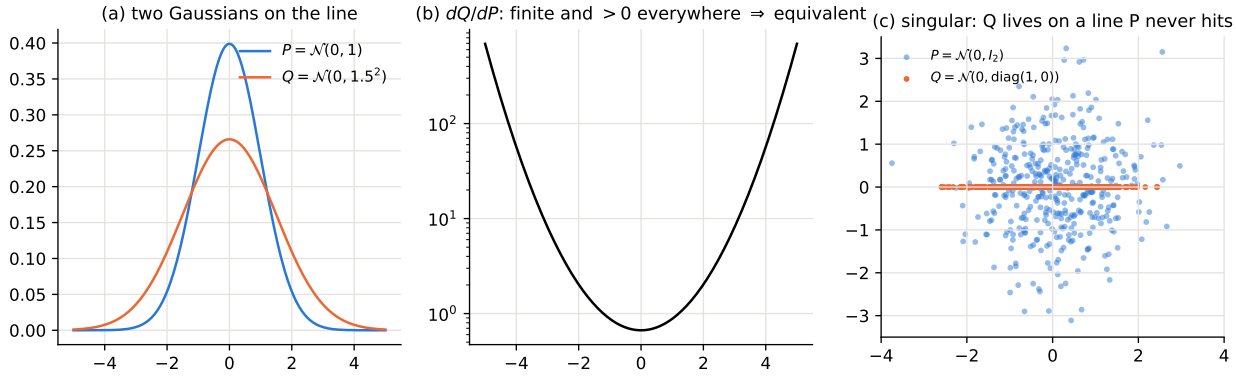



Figure 1: Finite dimensions. (a,b) Two non-degenerate Gaussians are equivalent: the density ratio is finite and positive everywhere. (c) A degenerate Gaussian sits on a line, which has probability 0 under $\mathcal{N}(0, I_2)$, so the two measures are singular.

So singularity is an *infinite-dimensional* phenomenon. On a function space, as on an ever-finer mesh, it arises without any degeneracy.

## 1.2 Infinitely many independent coordinates

Take $x = (x_1, x_2, \dots)$ with

$$P = \mathcal{N}(0, I), \qquad Q = \mathcal{N}\big(0, \operatorname{diag}(1 + t_j)\big), \qquad t_j > -1.$$

$t_j$ is the relative change of variance in coordinate $j$. Restricted to the first $n$ coordinates, both measures are non-degenerate Gaussians and the log density ratio of one sample is a sum of independent terms:

$$\ell_n(x) = \log \frac{dQ}{dP}(x) = \sum_{j=1}^{n} \Big[ -\tfrac{1}{2}\log(1 + t_j) + \tfrac{1}{2}\,\frac{t_j}{1 + t_j}\, x_j^2 \Big]. \tag{1}$$

**Mean and spread of the one-sample statistic.** Using $\mathbb{E}x_j^2 = 1$ under $P$, $\mathbb{E}x_j^2 = 1 + t_j$ under $Q$ and $\mathrm{Var}(x_j^2) = 2(\mathbb{E}x_j^2)^2$:

$$\mathbb{E}_Q\,\ell_n = \tfrac{1}{2}\sum_j \big(t_j - \log(1+t_j)\big) = \mathrm{KL}(Q\|P) \approx \tfrac{1}{4}\sum_j t_j^2,$$
$$\mathbb{E}_P\,\ell_n = -\tfrac{1}{2}\sum_j \Big(\log(1+t_j) - \tfrac{t_j}{1+t_j}\Big) = -\mathrm{KL}(P\|Q) \approx -\tfrac{1}{4}\sum_j t_j^2,$$
$$\mathrm{Var}_P\,\ell_n = \tfrac{1}{2}\sum_j \Big(\tfrac{t_j}{1+t_j}\Big)^2 \approx \tfrac{1}{2}\sum_j t_j^2,$$

where $\approx$ is for small $t_j$. The two histograms of $\ell_n$ (one for samples from $P$, one for samples from $Q$) are therefore separated by about $\frac{1}{2}\sum t_j^2$, while each has width about $\sqrt{\frac{1}{2}\sum t_j^2}$. Their separation in units of width is

$$\frac{\text{gap}}{\text{width}} \approx \sqrt{\tfrac{1}{2}\textstyle\sum_{j\le n} t_j^2}\,.$$

**Proposition 1.2** (Diagonal Feldman–Hájek)**.** *If $\sum_j t_j^2 = \infty$, one sample identifies the measure with certainty as $n \to \infty$, so $P \perp Q$. If $\sum_j t_j^2 < \infty$, the histograms keep a fixed overlap and $P \sim Q$.*

The rigorous version is **Kakutani's theorem** (1948). Products of equivalent measures $\bigotimes P_j$ and $\bigotimes Q_j$ are equivalent iff the product of Hellinger affinities $\prod_j \rho_j$ is positive, and otherwise singular. For one Gaussian coordinate

$$\rho_j = \int \sqrt{p_j q_j}\,dx = \Big(\frac{2\sqrt{1+t_j}}{2+t_j}\Big)^{1/2}, \qquad -\log\rho_j = \frac{t_j^2}{16} + O(t_j^3),$$

so $\prod \rho_j > 0 \iff \sum t_j^2 < \infty$. Note the *dichotomy*: there is no in-between. Gaussian measures are either equivalent or singular.

**i.i.d. case.** With $t_j = 0.2$ for all $j$, every coordinate is a harmless 20% change, yet $\sum t_j^2 = \infty$. Fig. 2 shows the best one-sample test, "say $Q$ iff $\ell_n > 0$", going from coin-flipping at $n = 1$ to never erring at $n = 10^4$. The reason is the law of large numbers: $\frac{1}{n}\sum x_j^2 \to 1$ under $P$ and $\to 1.2$ under $Q$. These two events are disjoint and carry probability one under their own measure.

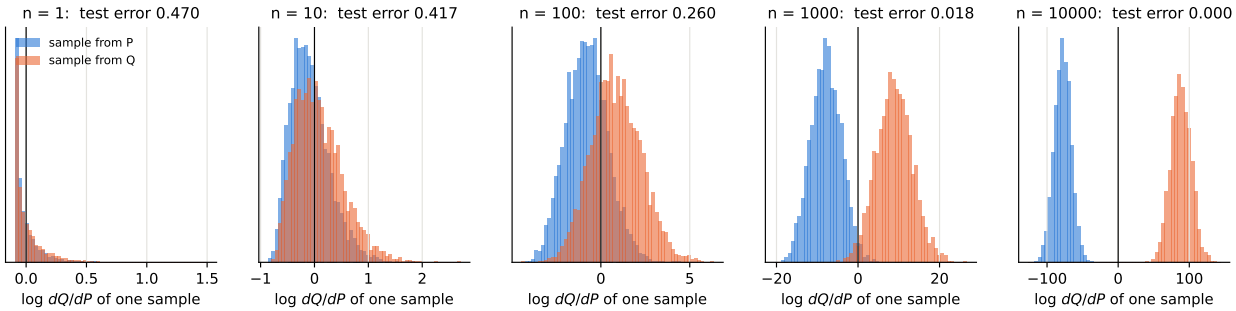



Figure 2: $t_j = 0.2$ in every coordinate. Histograms of the one-sample log likelihood ratio (1) under $P$ (blue) and $Q$ (orange). The title gives the error of the best test. As the number of coordinates grows the histograms separate completely: singular.

**Decaying perturbations.** Fig. 3 compares three sequences: $t_j = 0.2$ ($\sum t_j^2 = \infty$, fast), $t_j = j^{-1/4}$ ($\sum t_j^2 = \sum j^{-1/2} = \infty$, slow), and $t_j = 1/j$ ($\sum t_j^2 = \pi^2/6$). For $t_j = 1/j$ the affinity, the test error and the histograms all *freeze* as $n$ grows: extra coordinates bring no new evidence. That is what equivalence looks like.

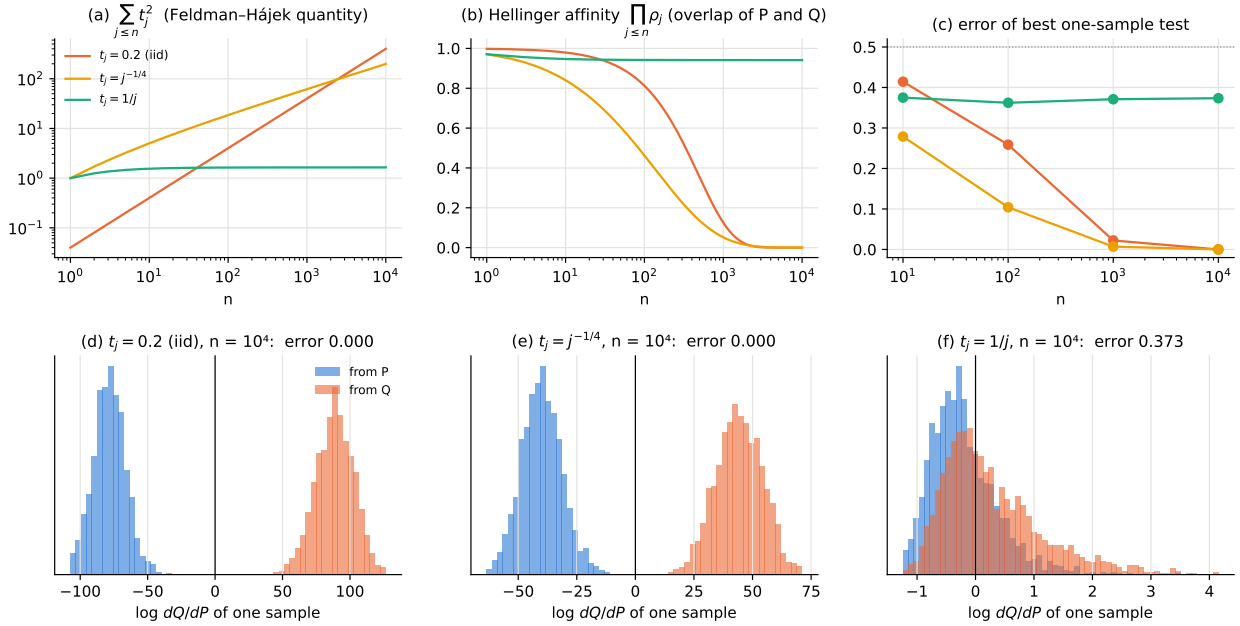



Figure 3: (a) The Feldman–Hájek quantity $\sum_{j\le n} t_j^2$. (b) The overlap of $P$ and $Q$ (Hellinger affinity). (c) The best one-sample test error. (d–f) Histograms at $n = 10^4$. Only $t_j = 1/j$, with a finite sum, keeps a permanent overlap.

### 1.3 Mean shifts and the Cameron–Martin space

So far $P$ and $Q$ had the same mean and different variances. Now keep the covariance and move the mean:

$$P = \mathcal{N}(0, I_n), \qquad Q = \mathcal{N}(m, I_n), \qquad m = (m_1, \dots, m_n).$$

Every $x_j$ still has variance 1; under $Q$ it is centred at $m_j$ instead of 0.

**The one-sample statistic.** Writing out the two Gaussian densities, the $x_j^2$ terms cancel and

$$\ell_n(x) = \log \frac{dQ}{dP}(x) = \sum_{j=1}^{n} \left( m_j x_j - \tfrac{1}{2} m_j^2 \right) = \langle m, x \rangle - \tfrac{1}{2}\|m\|^2.$$

This is a *matched filter*: project the sample onto the direction $m$ and compare with the midpoint. It is linear in $x$, so it is exactly Gaussian:

$$\begin{aligned} x \sim P: \quad & \langle m, x \rangle \sim \mathcal{N}(0, \|m\|^2) && \Rightarrow \quad \ell_n \sim \mathcal{N}\big(-\tfrac{1}{2}\|m\|^2,\ \|m\|^2\big), \\ x \sim Q: \quad & \langle m, x \rangle = \|m\|^2 + \langle m, z \rangle && \Rightarrow \quad \ell_n \sim \mathcal{N}\big(+\tfrac{1}{2}\|m\|^2,\ \|m\|^2\big). \end{aligned}$$

The two histograms are separated by $\|m\|^2$ and each has width $\|m\|$. Gap over width is $\|m\|$, and the best test (say $Q$ iff $\ell_n > 0$) has error $\Phi(-\|m\|/2)$.

**Geometric picture.** Along the direction $m/\|m\|$ the noise has standard deviation 1 (white noise has the same spread in every direction), while the signal has length $\|m\|$. The signal-to-noise ratio is $\|m\|$. Each coordinate contributes $m_j^2$ to $\|m\|^2$, so $m_j^2$ is the evidence coordinate $j$ supplies, just as $t_j^2$ was for variances.

$$P \sim Q \iff \|m\|^2 = \sum_j m_j^2 < \infty.$$

**Examples (Fig. 4).**

- $m_j = 1/j$: $\|m\|^2 \to \pi^2/6 \approx 1.64$, $\|m\| \approx 1.28$. The error freezes at $\Phi(-0.64) \approx 0.26$ however many coordinates are added: **equivalent**.
- $m_j = 1/\sqrt{j}$: $\|m\|^2 = \sum 1/j \approx \log n + 0.58 \to \infty$. The error decays to 0, but only logarithmically: at $n = 10^4$, $\|m\| \approx 3.1$ and the error is still $\approx 0.06$. Even so, it is **singular**. Each individual shift $m_j \to 0$, but not fast enough.

**When the coordinates are not white: the Cameron–Martin space.** For a general prior $P = \mathcal{N}(0, C)$ with eigenpairs $(\lambda_j, \phi_j)$, coordinate $j$ has noise standard deviation $\sqrt{\lambda_j}$. A shift $m_j = \langle m, \phi_j \rangle$ must be measured in units of that noise: whitening gives $\tilde{m}_j = m_j/\sqrt{\lambda_j}$, and

$$P \sim Q \iff \sum_j \frac{m_j^2}{\lambda_j} = \|C^{-1/2} m\|^2 < \infty \iff m \in \mathcal{H}_P := C^{1/2} H.$$

$\mathcal{H}_P$ is the *Cameron–Martin space* of $P$. Because $\lambda_j \to 0$ quickly for a smooth prior, fine-scale coordinates are expensive. A shift that has any appreciable fine-scale content is fatal, because the prior has almost no noise there to hide it in.

**For our SPDE prior.** $(\kappa^2 - \Delta)u = \xi$ has $\lambda_j = (\kappa^2 + \mu_j)^{-2}$, where the $\mu_j$ are the Dirichlet Laplacian eigenvalues. Then

$$\|C^{-1/2} m\|^2 = \sum_j (\kappa^2 + \mu_j)^2 m_j^2 = \|(\kappa^2 - \Delta) m\|_{L^2}^2,$$

so $\mathcal{H}_P = \{m : (\kappa^2 - \Delta)m \in L^2,\ m = 0 \text{ on } \partial\Omega\}$, the Sobolev space $H^2$ with boundary conditions. Two consequences follow.

- **A typical sample is *not* in its own Cameron–Martin space.** For a sample $u$, $(\kappa^2 - \Delta)u = \xi$ is white noise, and $\|\xi\|_{L^2}^2 = \sum \xi_j^2 = \infty$. So shifting the prior by one of its own typical draws gives a singular measure. The Cameron–Martin space is much smoother than the samples: it has probability zero under $P$.
- **A deterministic source.** A known forcing $q = \xi + a\, g$ shifts the mean by $m = a(\kappa^2 - \Delta)^{-1} g$. Then $\|C^{-1/2} m\| = |a|\, \|g\|_{L^2}$, so the shifted prior is equivalent iff $g \in L^2$. A smooth or extended source is fine. A *point* source $g = \delta_{x_0}$ is not in $L^2$ (in any dimension), so its mean shift is singular. On a mesh, $\delta$ becomes a spike of height $1/h^d$ at one node, and $\|g\|^2 \approx h^{-d} \to \infty$.

A *random* source amplitude ($a \sim \mathcal{N}(0, c)$) gives the covariance version of the same condition. In whitened coordinates the added eigenvalue is $t = c\, \|g\|_{L^2}^2$, finite iff $g \in L^2$ (on a mesh, $t = c\, h\, g^\top g$).

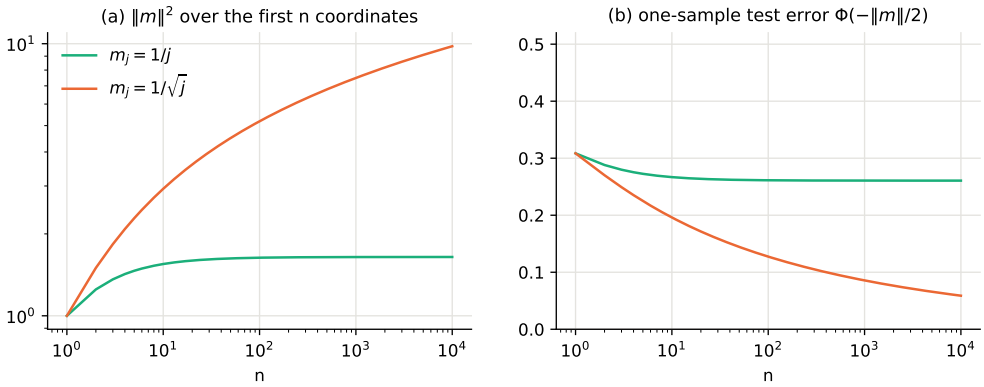



Figure 4: Mean shifts. $m_j = 1/j$ has finite $\|m\|$, so the measures are equivalent. $m_j = 1/\sqrt{j}$ has infinite $\|m\|$, so they are singular.

### 1.4 The Feldman–Hájek theorem

Remove the diagonal assumption. Let $H$ be a separable Hilbert space ($L^2(\Omega)$ for us), and $P = \mathcal{N}(m_0, C_0)$, $Q = \mathcal{N}(m_1, C_1)$ Gaussian measures with injective trace-class covariances.

**Theorem 1.3** (Feldman 1958, Hájek 1958)**.** *$P$ and $Q$ are either equivalent or mutually singular. They are equivalent if and only if*

*(i)* $C_0^{1/2}H = C_1^{1/2}H$ *(same Cameron–Martin space);*

*(ii)* $m_1 - m_0 \in C_0^{1/2}H$*;*

*(iii)* $T := C_0^{-1/2} C_1 C_0^{-1/2} - I$ *is a* Hilbert–Schmidt *operator, i.e.* $\|T\|_{HS}^2 = \sum_j t_j^2 < \infty$*, where* $t_j$ *are the eigenvalues of* $T$*.*

See Da Prato & Zabczyk (2014, Thm. 2.25) or Bogachev (1998, Thm. 2.7.2).

**How to read it: whitening.** Let $z = C_0^{-1/2}(u - m_0)$. Under $P$, $z$ is white noise, $\mathcal{N}(0, I)$. Under $Q$ (taking $m_1 = m_0$), $z \sim \mathcal{N}(0, I + T)$. In the eigenbasis of $T$ this is exactly the diagonal problem of the previous subsection, with the $t_j$ now the eigenvalues of $T$. Condition (iii) is $\sum t_j^2 < \infty$. Condition (ii) is $\sum m_j^2 < \infty$ in whitened coordinates. Condition (i) is the technical requirement that $T$ is well defined and $I + T$ is invertible, so no $t_j$ reaches $-1$.

**KL divergence.** When $P \sim Q$ (zero means),

$$\mathrm{KL}(Q\|P) = \tfrac{1}{2} \sum_j \big(t_j - \log(1 + t_j)\big) = \tfrac{1}{2}\big(\operatorname{tr} T - \log\det(I + T)\big), \tag{2}$$

which is finite exactly when $\sum t_j^2 < \infty$, since $t - \log(1 + t) \approx t^2/2$. If $P \perp Q$, then $\mathrm{KL} = \infty$.

**The mesh-refinement test.** On an $N$-node mesh everything is finite. Compute the $N \times N$ matrix $T_N$ and watch $\|T_N\|_F$ and $\mathrm{KL}_N$ as $N$ grows. A plateau means $P \sim Q$. Growth without bound means $P \perp Q$.

### 1.5 Random functions: range versus amplitude

Take the 1D version of the reference prior on $[0,1]$ with $u(0) = u(1) = 0$:

$$(\kappa^2 - \partial_x^2)\, u = \tau\, \xi.$$

The sine functions $\phi_j(x) = \sqrt{2}\sin(j\pi x)$ are eigenfunctions of $-\partial_x^2$ with $\lambda_j = (j\pi)^2$. Hence

$$u = \sum_j c_j\, \phi_j, \qquad c_j \sim \mathcal{N}\Big(0, \frac{\tau^2}{(\kappa^2 + \lambda_j)^2}\Big) \text{ independent.}$$

Two such priors are diagonal in the same basis, with $1 + t_j = \mathrm{var}_Q(c_j)/\mathrm{var}_P(c_j)$.

- **Range change** $\kappa_0 \to \kappa_1$ (with $\tau$ fixed): $t_j = \big(\frac{\kappa_0^2+\lambda_j}{\kappa_1^2+\lambda_j}\big)^2 - 1 \approx \frac{2(\kappa_0^2-\kappa_1^2)}{\pi^2 j^2}$. So $\sum t_j^2 < \infty$: **equivalent**.
- **Amplitude change** $\tau_0 \to \tau_1$: $t_j = \tau_1^2/\tau_0^2 - 1$ for every $j$. This is the i.i.d. case: **singular**.

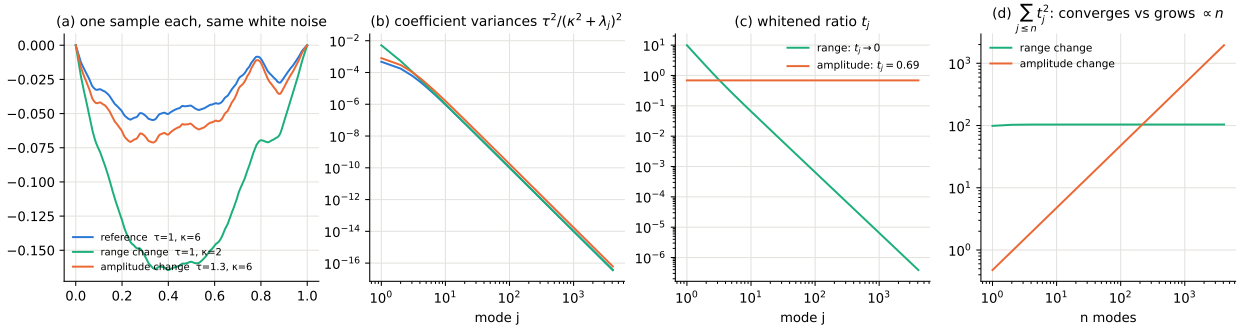



Figure 5: (a) One sample from the reference ($\tau = 1, \kappa = 6$), a range change ($\kappa = 2$) and an amplitude change ($\tau = 1.3$), all driven by the same white noise. (b) Coefficient variances: the range change merges with the reference at high modes, while the amplitude change stays parallel forever. (c) $t_j$. (d) $\sum t_j^2$ converges (to $\approx 104$, dominated by the first mode) for the range change and grows linearly for the amplitude change.

**The surprise (Fig. 5a).** The range change looks the most different to the eye, with bigger and smoother excursions, yet it is the *equivalent* one. The amplitude change looks almost like the reference, yet it is *singular*. Equivalence is decided by the **fine scales**, because there are infinitely many of them. A large-scale difference lives in finitely many modes, however dramatic it looks, and a single sample can never confirm it with certainty.

**Higher dimensions.** In $d$ dimensions the number of modes with $\lambda \le \Lambda$ grows like $\Lambda^{d/2}$, and for a range change $t \sim 1/\lambda$. Then $\sum t^2 \sim \int^\infty \Lambda^{d/2-1}\Lambda^{-2}\, d\Lambda < \infty$ iff $d < 4$. Range changes are harmless in $d \le 3$, which is Bolin & Kirchner's Cor. 3.3.

**The flip side: what one sample can teach you (microergodicity).** Fit $(\tau, \kappa)$ by maximum likelihood to *one* sample resolved to $N$ modes. The Fisher information is

$$I_\tau(N) = \frac{2N}{\tau^2} \to \infty, \qquad I_\kappa(N) = \frac{1}{2} \sum_{j \le N} \left( \frac{4\kappa}{\kappa^2 + \lambda_j} \right)^2 \to \text{const} < \infty.$$

$\hat{\tau}$ becomes exact as the sample is refined, while $\hat{\kappa}$ never settles (Fig. 6). In the notebook the spread of $\hat{\tau}$ falls from 0.26 to 0.01 between $N = 16$ and 4096, while the spread of $\hat{\kappa}$ stays near 2.5. This is the same fact as before, seen from the other side: parameters that change the measure's class are learnable from one sample, and parameters that keep it equivalent are not. $\tau$ (more precisely $\sigma^2\kappa^{2\nu}$ in Matérn form) is the *microergodic* parameter (Zhang 2004).

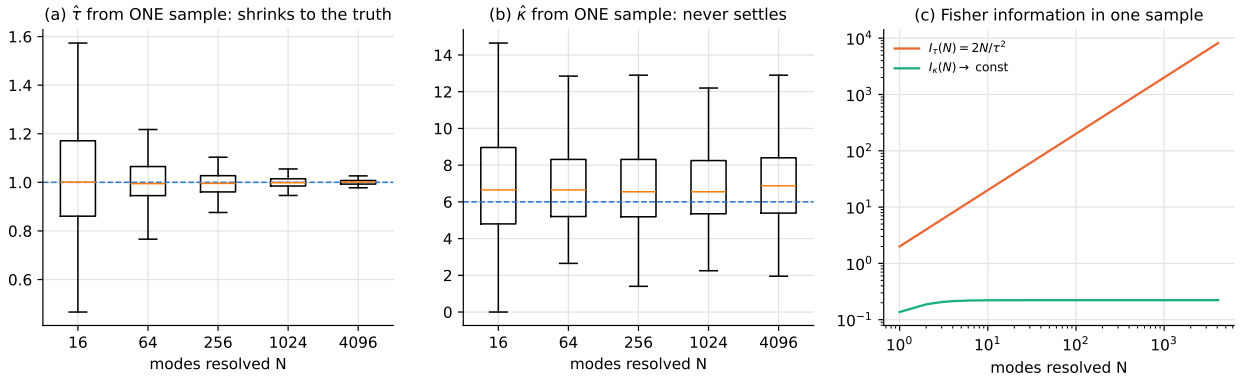



Figure 6: Maximum-likelihood estimates from a single sample (300 replicates, truth $\tau = 1, \kappa = 6$). (a) $\hat{\tau}$ converges as the resolution grows. (b) $\hat{\kappa}$ does not. (c) Fisher information: unbounded for $\tau$, bounded for $\kappa$.

**A practical consequence.** Because $\kappa$ costs almost nothing in likelihood, it is a cheap knob for a fit to turn.

## 1.6 Take-aways

1. Equivalence means the same null sets, which means a density $dQ/dP$ exists, which means one perfect sample cannot tell the measures apart.

2. Diagonal case: equivalent iff $\sum t_j^2 < \infty$ (variances) and $\sum m_j^2 < \infty$ (means). Feldman–Hájek is the same statement without a basis: $T$ is Hilbert–Schmidt and the mean lies in the Cameron–Martin space.

3. The decision is made at the **fine scales**. Large-scale changes (range $\kappa$, smooth sources, lower-order terms) are harmless however big they look. Changes that persist down to the smallest scales (amplitude $\tau$, the leading coefficient) are fatal however small they look.

4. Practically: what a fine sample determines exactly ($\tau$) cannot be "corrected" with finitely many directions, and what it never determines ($\kappa$) is a cheap knob a fit can slide along.

*Check yourself.* 1. Is $t_j = 1/\sqrt{j}$ equivalent or singular? Compute $\sum t_j^2$.

2. A constant diffusion coefficient $K \equiv 2$: equivalent or singular? (Hint: $-\nabla \cdot (2\nabla u) + \kappa^2 u = \xi$ is $2\big(\frac{\kappa^2}{2} - \Delta\big)u = \xi$. Which parameter of §1.5 changed?)

3. A source shape $g$ must be in $L^2$, so that its response $(\kappa^2 - \Delta)^{-1} g$ is in $H^2$. What happens to $t = c\, h\, g^\top g$ when $g$ becomes a delta at the grid scale ($g_i = 1/h$ at one node)?

**Next chapter (planned).** Whitening and the operator $T$ in two dimensions, and the symbols of differential operators.

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